Chapter 14

14.1

Data on satellite fading at L-band in various environments can be found in “Handbook of Propagation Effects for Vehicular and Personal Mobile Satellite Systems”, by J. Goldhirsh and W. Vogel, and in “Satellite Systems for Personal and Broadband Communications”, by E. Lutz, M. Werner, and A. Jahn, as well as from various other sources. Typically, one can expect fading levels of 15 to 20 dB for domestic and commercial buildings, for 95% link availability. For vehicles, the fading levels can be 20 dB or more. On the other hand, a line-of-sight system (as when the handset is used outdoors with little or no blockage to the satellite) would require a link margin of 0 dB in principal, although a few dB of margin would provide a more robust system. In view of this data, it is not clear why the Iridium system was designed with a link margin of 16 dB.

14.2

$$F_\theta(\theta, \phi) = A \sin^2\theta \cos\phi$$

Main beam at $\theta = 90°$, $\phi = 0°$ or $180°$

3 dB points at $\theta = 90°$ (az) plane:

$$\cos\phi = 0.707 \Rightarrow \phi = 45° \text{ or } 135°$$

$$HPBW_\phi = 135 - 45 = 90° \quad ✓$$

3 dB points in $\phi = 0°$ (el) plane:

$$\sin^2\theta = 0.707 \Rightarrow \theta = \sin^{-1}\sqrt{.707} = 57.2°,\ 122.8°$$

$$HPBW_\theta = 122.8 - 57.2 = 65.6°$$

$$\int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} F^2(\theta,\phi)\sin\theta\, d\theta\, d\phi = \int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} \sin^5\theta \cos^2\phi\, d\theta\, d\phi$$

$$= \left(\frac{16}{15}\right)\pi$$

$$D = \frac{4\pi F_{MAX}^2}{\iint} = \frac{15}{4} = 3.75 = \underline{5.74\ dB}$$

linearly polarized in vertical direction.

14.3 $F_\theta(\theta,\phi) = \begin{cases} A\sin\theta & \text{for } 0 \le \theta \le \pi/2 \\ 0 & \text{otherwise} \end{cases}$

$$D = \frac{4\pi F_{MAX}^2}{\int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} F_\theta^2(\theta,\phi)\sin\theta\, d\theta\, d\phi} = \frac{4\pi}{2\pi\int_{\theta=0}^{\pi/2}\sin^3\theta\, d\theta} = \frac{4\pi}{2\pi\left(\frac{2}{3}\right)}$$

$= 3 = \underline{\underline{4.8\,dB}}$ (verified with PCAAD)

14.4 $f = 12.4\,GHz$, $Diam = 18'' = 0.457\,m$, $\eta_{ap} = 65\%$

$\lambda = \frac{c}{f} = 0.0242\,m$ ✓

$A = \pi R^2 = \pi\left(\frac{Diam}{2}\right)^2 = 0.164\,m^2$

From (14.13)

$$D = \eta_{ap}\frac{4\pi A}{\lambda^2} = (0.65)\frac{4\pi(0.164)}{(0.0242)^2} = 2287 = \underline{\underline{33.6\,dB}}$$ ✓

14.5 $f = 38\,\text{GHz}$, $G = 39.0\,\text{dB}$, $\text{Diam} = 12.0''$, $\eta_{rad} = 90\%$

a) From (14.11), $D = \dfrac{G}{\eta_{rad}} = \dfrac{10^{39/10}}{0.9} = 8{,}826.$

From (14.13), $D = \dfrac{4\pi A}{\lambda^2}\eta_{ap}$

$$\eta_{ap} = \frac{\lambda^2 D}{4\pi A} = \left(\frac{c}{\pi f\,\text{Diam}}\right)^2 D = \underline{\underline{60\%}}$$

b) From (14.9), $D = \dfrac{32{,}400}{\theta_3^2}$

$$\theta_3 = \sqrt{\frac{32{,}400}{8826}} = \underline{\underline{1.9^\circ}}$$

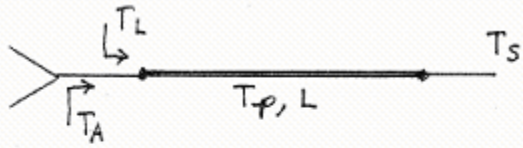
14.6 From (14.18), $T_A = \eta_{rad} T_b + (1-\eta_{rad}) T_p$

$$= (T_b - T_p)\eta_{rad} + T_p$$

Thus,

$$\eta_{rad} = \frac{T_A - T_p}{T_b - T_p} = \frac{105-290}{5-290} = \underline{65\%}$$

14.7



$T_A = \eta_{rad} T_b + (1-\eta_{rad}) T_p$ (at antenna) (14.18)

$T_L = (L-1) T_p$ (at antenna) (10.15)

$T_S = \frac{1}{L}(T_A + T_L)$ (at output)

$= \frac{1}{L}[\eta_{rad} T_b + (1-\eta_{rad}) T_p] + \frac{(L-1)}{L} T_p$ (14.20) ✓

14.8

$$T = \eta_{rad} T_b + (1-\eta_{rad}) T_p + T_R$$

$T_b = 50K$
$T_p = 290K$
$F = 1.1dB = 1.29$
$L = 2.5dB = 1.78$
$G = 33.5dB = 2240$

The noise temperature of the receiver is,

$$T_R = (F-1)T_0 = (1.29-1)(290) = 84K$$

The efficiency of the array is,

$$\eta_{rad} = \frac{1}{L} = \frac{1}{1.78} = 0.56$$

Thus,

$$T = (0.56)(50) + (1-0.56)(290) + 84 = \underline{240K}$$

Then,

$$\frac{G}{T}(dB) = 10\log\frac{2240}{240} = \underline{9.7\ dB/K}$$

This value is well below the desired minimum of 12 dB/K.

14.9 Solving (14.23) for G gives,

$$G = \frac{4\pi S R^2}{P_t} = \frac{4\pi(7.5\times10^{-3})(300)^2}{85} = 100 = \underline{20\ dB}$$

14.10

1) RADIO LINK: $f = 28\,GHz \Rightarrow \lambda = 0.0107\,m$; $G_t = G_r = 25\,dB = 316$.

Let $P_r = 1\,W$. Then,

$$P_t = \frac{(4\pi R)^2}{P_r G_t G_r \lambda^2} = \frac{(4\pi)^2 (5000)^2}{(1)(316)^2 (0.0107)^2} = 3.45 \times 10^8\,W$$

$$\text{ATTENUATION} = 10 \log \frac{P_r}{P_t} = 10 \log \frac{1}{3.45 \times 10^8} = -\underline{85.4\,dB} \checkmark$$

2) WIRED LINK: $\alpha = 0.05\,dB/m = 0.0057$ neper/m; $4 \times 30\,dB$ REPEATERS.

$$\text{ATTENUATION OF LINE} = 10 \log e^{-2\alpha R}$$
$$= 10 \log e^{-2(0.0057)(5000)}$$
$$= -250\,dB$$

$$\text{TOTAL LOSS} = -250 + 4(30) = \underline{-130\,dB} \checkmark$$

The radio link has much less link loss than the wired link, and will thus require less transmit power.

14.11 GSM downlink, 935-960 MHz, EIRP = 20 W, $G_r = 0\,dBi$, $T = 450\,K$, $SNR = 10\,dB$, $LM = 30\,dB$, $B = 200\,kHz$, $F_R = 8\,dB$.

$f = 947.5\,MHz \Rightarrow \lambda = 0.317\,m$, $G_r = 0\,dBi = 1$

$F_R = 8\,dB = 6.31$, $SNR = 10\,dB = 10$.

$$T_{SYS} = T_A + T_R = T_A + (F_R - 1)T_0 = 450 + (6.31-1)(290)$$
$$= 1990\,K$$
$$N_0 = kT_{SYS}B = (1.38\times10^{-23})(1990)(200\times10^{3})$$
$$= 5.5\times10^{-15}\,W = -112.6\,dBm \quad \text{(at rec. input)}$$

$$S_0(dBm) = \left(\frac{S_0}{N_0}\right) + N_0 + LM = 10 - 112.6 + 30 = -72.6\,dBm$$
$$= 5.5\times10^{-11}\,W$$

$$R = \sqrt{\frac{P_t G_t G_r \lambda^2}{(4\pi)^2 S_0}} = \sqrt{\frac{(20)(1)(.317)^2}{(4\pi)^2(5.5\times10^{-11})}} = \underline{15.2\,km}$$

(stated base station range for GSM 900 is 2-20 km)

14.12 Carrier power at receiver:

$$C = S_i G_A G/L \qquad (S_i \text{ ref. to antenna w/ } 0\,dB_i)$$

at input to amplifier:

$$T_e = T_A + (F-1)T_0 + (L-1)T_0/G$$

The noise power at input to receiver:

$$N = kT_e G/L = \frac{C}{(C/N)}$$

$L = 25\,dB = 316.2$
$S_i = 1\times10^{-16}$
$G_A = 5\,dB = 3.16$
$\frac{C}{N} = 32\,dB = 1580.$
$G = 10\,dB = 10$

So,

$$T_e = \frac{CL}{k\left(\frac{C}{N}\right)G} = \frac{S_i G_A}{k\left(\frac{C}{N}\right)}$$

$$F = 1 + \frac{T_e}{T_0} - \frac{T_A}{T_0} - \frac{(L-1)}{G} = 1 + \frac{S_i G_A}{kT_0\left(\frac{C}{N}\right)} - \frac{T_A}{T_0} - \frac{(L-1)}{G}$$

$$= 1 + \frac{(1\times10^{-16})(3.16)}{(1.38\times10^{-23})(290)(1.58\times10^{3})} - \frac{300}{290} - \frac{(316.2-1)}{10}$$

$$= 18.4 = \underline{12.6\,dB}$$

14.13 $N_0 = kT_bB = S_0$ for $S_0/N_0 = 0\,dB$

$$kT_bB = P_r = \frac{P_t G^2 \lambda^2}{(4\pi R)^2}$$

$$R = \sqrt{\frac{P_t G^2 \lambda^2}{16\pi^2 k T_b B}} = \sqrt{\frac{(1000)(2.51)^2(4.48)^2}{16\pi^2(1.38\times10^{-23})(4)(4\times10^{6})}} = \underline{1.9\times10^{9}\,m}$$

If $SNR = 30\,dB = 1000$, $R = 6.0\times10^{7}\,m$

$R_{VENUS} = 4.2\times10^{10}\,m$, so the signal will not even reach the nearest planet.

14.14 Mariner 10, PSK, $P_b = 0.05$, $(E_b/\eta_0) = 1.4\,dB$,
$R = 1.6 \times 10^8\,km$, $f = 2.295\,GHz$. $G_t = 27.6\,dB$
$P_t = 16.8\,W$, $G_r = 61.3\,dB$, $T_{sys} = 13.5\,K$

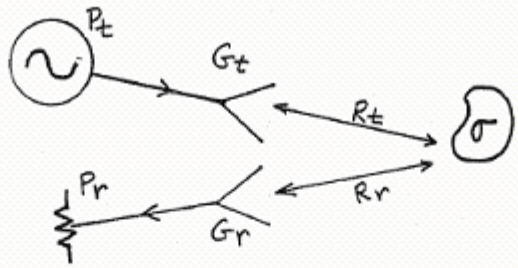
$\lambda = 0.1307\,m$, $P_t = 12.25\,dBW$, $E_b/\eta_0 = 1.38$

$L_0 = 20 \log(4\pi R/\lambda) = 263.7\,dB$.
$P_r = P_t + G_t - L_0 + G_r = -162.6\,dBW = 5.56 \times 10^{-17}\,W$ ✓
$\eta_0 = kT_{sys}$

From (14.36), $R_b = \left(\frac{P_r}{\eta_0}\right)\left(\frac{\eta_0}{E_b}\right) = \frac{5.56 \times 10^{-17}}{(1.38 \times 10^{-23})(13.5)}\left(\frac{1}{1.38}\right)$

$= \underline{216\,kbps}$ ✓

14.15



From (14.23) the power density incident on the target is,

$$S = \frac{P_t G_t}{4\pi R_t^2}$$

The scattered power density at the receiver is, from (14.40),

$$S_r = \frac{P_t G_t \sigma(\theta_t, \phi_t; \theta_r, \phi_r)}{(4\pi)^2 R_t^2 R_r^2},$$

where $\sigma(\theta_t, \phi_t; \theta_r, \phi_r)$ is the radar cross-section of the target seen at θ_r, ϕ_r with an incident wave at θ_t, ϕ_t. Then the received power can be found using (14.14):

$$P_r = P_t \frac{G_t G_r \lambda^2 \sigma(\theta_t, \phi_t; \theta_r, \phi_r)}{(4\pi)^3 R_t^2 R_r^2} \quad \checkmark$$

14.16

When a pulse is transmitted at $t=0$, the return pulse must come back before the next pulse is transmitted at $t = T_r$, to avoid an ambiguity in range. The round-trip time for a pulse return is,

$$T = 2R/c,$$

so the maximum unambiguous range is,

$$R_{MAX} = \frac{cT_r}{2} = \frac{c}{2f_r} \quad \checkmark$$

14.17 From (14.43) the doppler frequencies are,

$$f_{d(MIN)} = \frac{2 v_{MIN} f_o}{c} = \frac{2(1\frac{m}{sec})(12GHz)}{3\times10^8 m/sec} = 80\ Hz$$

$$f_{d(MAX)} = \frac{2 v_{MAX} f_o}{c} = \frac{2(20\ m/sec)(12GHz)}{3\times10^8\ m/sec} = 1.6\ kHz$$

So the necessary passband is 80 – 1600 Hz.

14.18 From (14.41) the received power is,

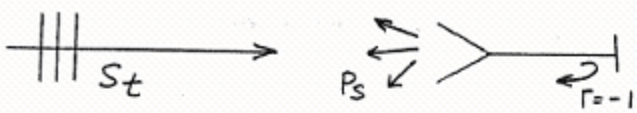
$$P_r = P_t \frac{G^2\lambda^2\sigma}{(4\pi)^3 R^4} = 1000\frac{(1000)^2(0.15)^2(20)}{(4\pi)^3(10^4)^4} = 2.27\times10^{-11}\ W$$

$$= -76\ dBm$$

The transmitter power is $10\log(10^4)(10^3) = 70\ dBm$. So the isolation between receiver and transmitter must be,

$$I = 70\ dBm - (-76\ dBm) + 10\ dB = 156\ dB$$

14.19



Assume an incident plane wave with power density S_t. Then the received power of the antenna is, from (13.14) and (14.15)

$$P_\ell = S_t A_e = S_t \frac{\lambda^2 G}{4\pi}$$

Because of the short-circuit termination, all of this power is re-transmitted (assuming a lossless antenna), giving a radiated power in the main beam direction of, $P_s = G P_\ell$. Then the RCS can be found from (14.39):

$$\sigma = \frac{P_s}{S_t} = \frac{\lambda^2 G^2}{4\pi} \checkmark$$

14.20

$$\Delta T_{TRUE} = T_p - T_2$$

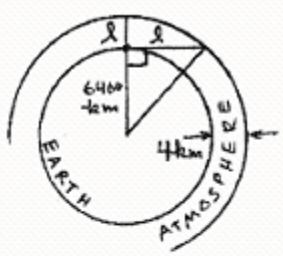
$$\Delta T_s = T_s\Big|_{T_B=T_1} - T_s\Big|_{T_B=T_2}$$

From (14.19), with $L=1$, $T_s = (1-\Gamma^2)\left[\eta_{rad} T_b + (1-\eta_{rad}) T_p\right]$

$$\Delta T_s = (1-\Gamma^2)\left[\cancel{\eta_{rad} T_p} + \cancel{(1-\eta_{rad})} T_p\right] - (1-\Gamma^2)\left[\eta_{rad} T_2 + \cancel{(1-\eta_{rad})} T_p\right]$$

$$\frac{\Delta T_s}{\Delta T_{TRUE}} = \frac{(1-\Gamma^2)\eta_{rad}(T_p - T_2)}{T_p - T_2} = (1-\Gamma^2)\eta_{rad} \quad \checkmark$$

14.21

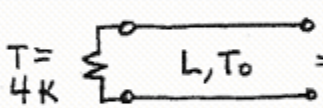


LOOKING TOWARD ZENITH, $\ell = 4000$ m. $= 4$ km.

LOOKING TOWARD HORIZON,

$$\ell = \sqrt{(6404)^2 - (6400)^2} = 226 \text{ km}.$$

$\alpha = 0.005$ dB/km



$$\Rightarrow T_e = \frac{4}{L} + (L-1)T_o$$

AT ZENITH: $L = (0.005 \text{ dB/km})(4 \text{ km}) = 0.02 \text{ dB} = 1.0046$

$T_e = \underline{5.3K}$

AT HORIZON: $L = (0.005 \text{ dB/km})(226 \text{ km}) = 1.13 \text{ dB} = 1.297$

$T_e = \underline{89K}$

14.22 $f = 28\,GHz \Rightarrow \lambda_0 = 0.0107\,m,\ G = 32\,dB = 1585,\ P_t = 5\,W$

a) $R = \sqrt{\dfrac{G P_t}{4\pi S}} = \sqrt{\dfrac{(1585)(5)}{4\pi(.01\,W/cm^2)}} = 251\,cm = \underline{2.51\,m}$

b) $G = 32 - 10 = 22\,dB = 158.5$

$R = \sqrt{\dfrac{(158.5)(5)}{4\pi\,(.01\,W/cm^2)}} = 79.5\,cm = \underline{0.8\,m}$

c) $d = \sqrt{\dfrac{\lambda^2 D}{\pi^2}} = 0.175\,m$ (use $D = \dfrac{G}{\eta_{ap}} = 2641$)

$R_{ff} = \dfrac{2d^2}{\lambda} = \underline{5.7\,m}$

(neither distance is in the far-field of the antenna)

14.23

$S = 1300\ W/m^2 = \frac{1}{2}|\bar{E}||\bar{H}| = \frac{1}{2\eta_0}|\bar{E}|^2 = \frac{\eta_0}{2}|\bar{H}|^2$

$|\bar{E}| = \sqrt{2\eta_0 S} = \sqrt{2(377)(1300)} = 990\ V/m$

$|\bar{H}| = \sqrt{\dfrac{2S}{\eta_0}} = \sqrt{\dfrac{2(1300)}{377}} = 2.6\ A/m$

check: $\frac{1}{2}|\bar{E}||\bar{H}| = \frac{1}{2}(990)(2.6) = 1300\ W/m^2$ ✓

6.26 The unperturbed TE_{101} cavity fields are,

$$E_y = A \sin\frac{\pi x}{a} \sin\frac{\pi z}{d}$$

$$H_x = \frac{-jA}{Z} \sin\frac{\pi x}{a} \cos\frac{\pi z}{d} \qquad ; \; Z = k\eta/\beta$$

$$H_z = \frac{j\pi A}{k\eta a} \cos\frac{\pi x}{a} \sin\frac{\pi z}{d}$$

Then the numerator in (6.95) is,

$$\int_{v_0} (\Delta\epsilon |\bar{E}_0|^2 + \Delta\mu |\bar{H}_0|^2) dv = (\mu_r - 1)\mu_0 \int_{x=0}^{a} \int_{y=0}^{b} \int_{z=0}^{t} (|H_x|^2 + |H_z|^2) dz\, dy\, dx$$

$$= \mu_0(\mu_r - 1)\frac{ab}{2} A^2 \int_{z=0}^{t} \left(\frac{1}{Z^2} \cos^2\frac{\pi z}{d} + \frac{\pi^2}{k^2\eta^2 a^2} \sin^2\frac{\pi z}{d} \right) dz$$

$$= \mu_0(\mu_r - 1)\frac{ab}{2} A^2 \left[\frac{1}{Z^2}\left(\frac{z}{2} + \frac{\sin\frac{2\pi z}{d}}{4\pi/d} \right)\Bigg|_0^t + \frac{\pi^2}{k^2\eta^2 a^2}\left(\frac{z}{2} - \frac{\sin\frac{2\pi z}{d}}{4\pi/d} \right)\Bigg|_0^t \right]$$

$$= \mu_0(\mu_r - 1)\frac{ab}{2} A^2 \left[\frac{t}{2\eta^2} + \frac{\beta^2 - \pi^2/a^2}{k^2\eta^2} \frac{d}{4\pi} \sin\frac{2\pi t}{d} \right]$$

The denominator in (6.95) is $\frac{abd\,\epsilon_0 A^2}{2}$, so

$$\frac{\omega - \omega_0}{\omega_0} = \frac{-(\mu_r - 1)ab\,\eta^2 [\bullet]}{abd}$$

$$= \frac{-(\mu_r - 1)}{d} \left(\frac{t}{2} + \frac{\beta^2 - \pi^2/a^2}{k^2} \frac{d}{4\pi} \sin\frac{2\pi t}{d} \right)$$

For $t \ll d$ this simplifies to,

$$\frac{\omega - \omega_0}{\omega_0} \simeq -(\mu_r - 1)\left(\frac{t}{d}\right)\left(\frac{\beta^2}{k^2}\right)$$

6.27 Following Example 6.8:

at $x = a/2,\ z = 0$: $E_y = 0$

$H_x = \frac{-jA}{Z}$, $Z = k_0\eta_0/\beta$

$H_y = 0$

Then,

$$\int_{\Delta v} (\mu|\bar{H}_0|^2 - \epsilon|\bar{E}_0|^2)dv = \mu_0 \frac{A^2}{Z^2}\Delta V \quad ; \quad \Delta V = \pi l r_0^2$$

$$\int_{\Delta v} (\mu|\bar{H}_0|^2 + \epsilon|\bar{E}_0|^2)dv = \frac{V_0 \epsilon_0 A^2}{2}$$

So (6.102) reduces to,

$$\frac{\omega - \omega_0}{\omega_0} = \frac{2\mu_0 \Delta V}{Z^2 \epsilon_0 V_0} = \frac{2\eta_0^2 \Delta V \beta^2}{k_0^2 \eta_0^2 V_0} = \frac{2\beta^2}{k_0^2}\frac{\Delta V}{V_0}$$

(an increase in resonant frequency)